

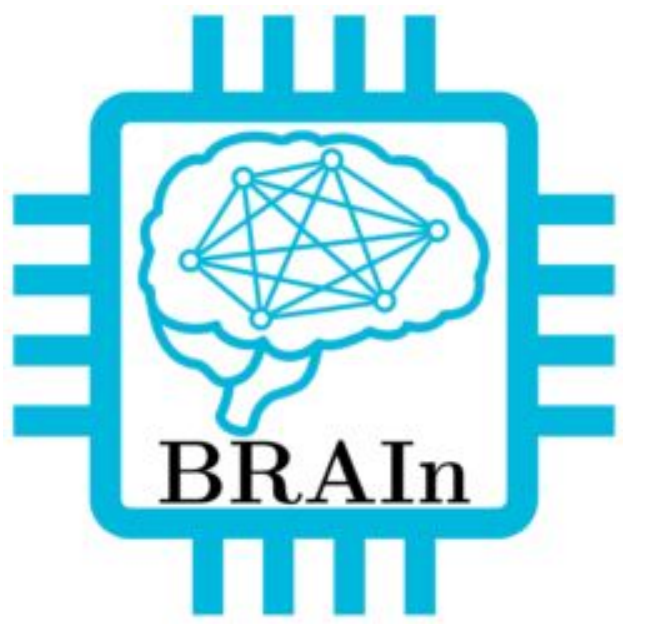


IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Unsupervised Adaptive Deep Learning Method For BCI Motor Imagery Decoding

Yassine El Ouahidi, Giulia Lioi, Nicolas Farrugia, Bastien Padeloup and Vincent Gripon

IMT Atlantique, Lab-STICC, UMR CNRS 6285, Brest, France



Introduction

- We propose an unsupervised adaptive method, suitable for real-time BCI
- SOTA on Cross-Subject setups on Motor Imagery decoding datasets
- Designed for Deep Learning models, without having to retrain the model
- We bridge the gap between offline and online BCI performance

Introduction

Motivation

- ❖ Brain Computer Interfaces (BCI) systems are promising for many applications, such as Motor Imagery (MI) decoding;
- ❖ Real-time application of MI-based BCIs is challenging due to EEG signal variability and non-stationarity;
- ❖ Our Goal is to develop an adaptive method that maintains performance of a system without requiring model retraining.

BCI Learning setups

- ❖ **Offline** setup allows for the use of **all trials** after they have been collected, offering high accuracy but not necessarily being suitable for real-time use;
- ❖ **Online** setup is designed for **real-time** applications, inducing some performance trade-offs.
- ❖ **Adaptive** setup proposes leveraging **incoming trials** as they arrive, making it suitable for real-time BCI. (*supervised* if the incoming trials labels are accessible, *unsupervised* otherwise)

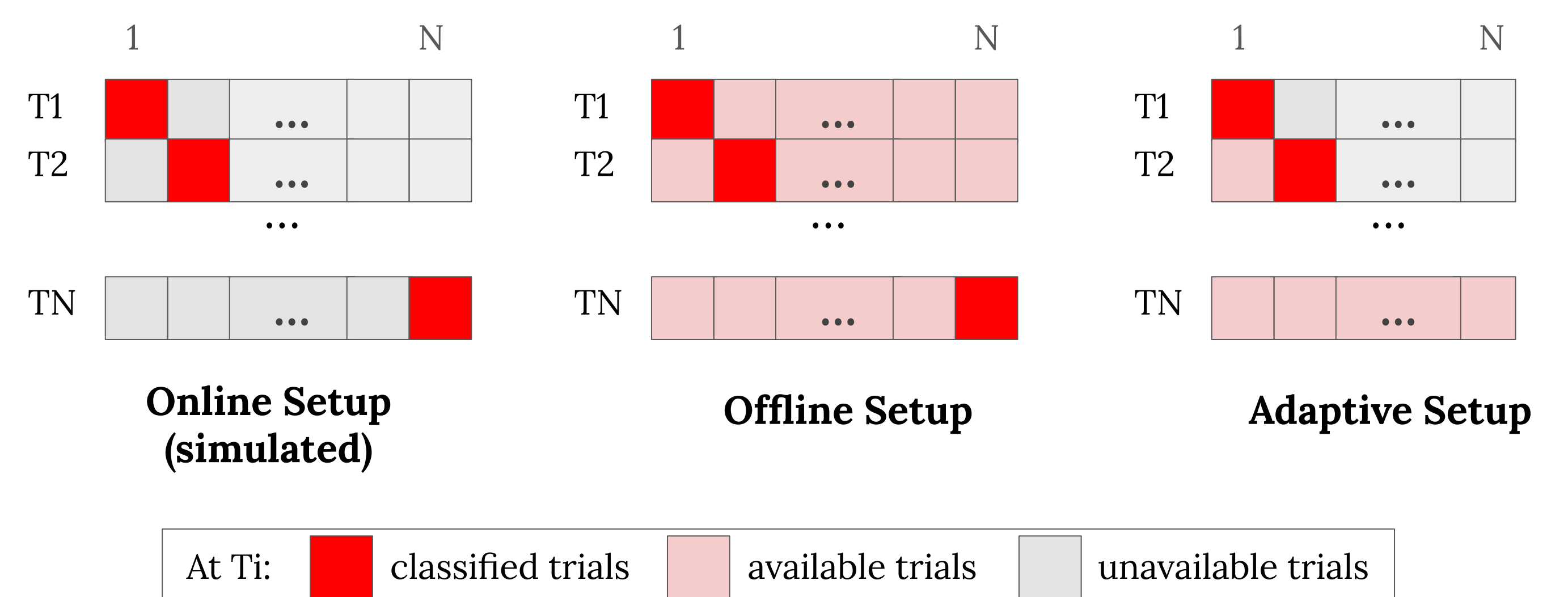
Method

We use EEGSimpleConv [1], a simple yet state-of-the-art 1D CNN, along with its training routine. We leverage two components of its routine:

- ❖ **Euclidean Alignment (EA):** Aligns each trial across subjects/sessions to a domain-invariant space using the arithmetic mean of covariance matrices.
- ❖ **Batch Normalization (BN) Trick:** Normalizes within the latent space of the model using statistics from evaluated trials, rather than the training set.

In our method, **EA** and **BN** statistics are updated in an **adaptive** way (after each incoming trial).

Setups differences, for an evaluated session with N Trials



Results

Results

Performance (% accuracy) Cross Subject

	BNCI	BNCI2	Large
	Cross-Subject		
Online	56.9	75.1	77.5
Adaptive	72.1	84.9	86.5
Adaptive + Buffer	72.2	84.2	88.3
Offline	72.7	85.1	89.3

Performance (% accuracy) Cross Subject +FT

	BNCI	BNCI2	Large
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Online	78.8	89.5	91.2
Adaptive	84.6	90.7	91.9
Adaptive + Buffer	85.9	91.7	91.9
Adaptive + Soft K-means	84.6	90.6	91.8
Offline	86.2	92.2	92.3

Comparison to the litterature

	RCT [12] ¹	Wimpff et al. [13]	Ours
Accuracy (%)	44.6	67.3	72.2

¹ Results reproduced by us

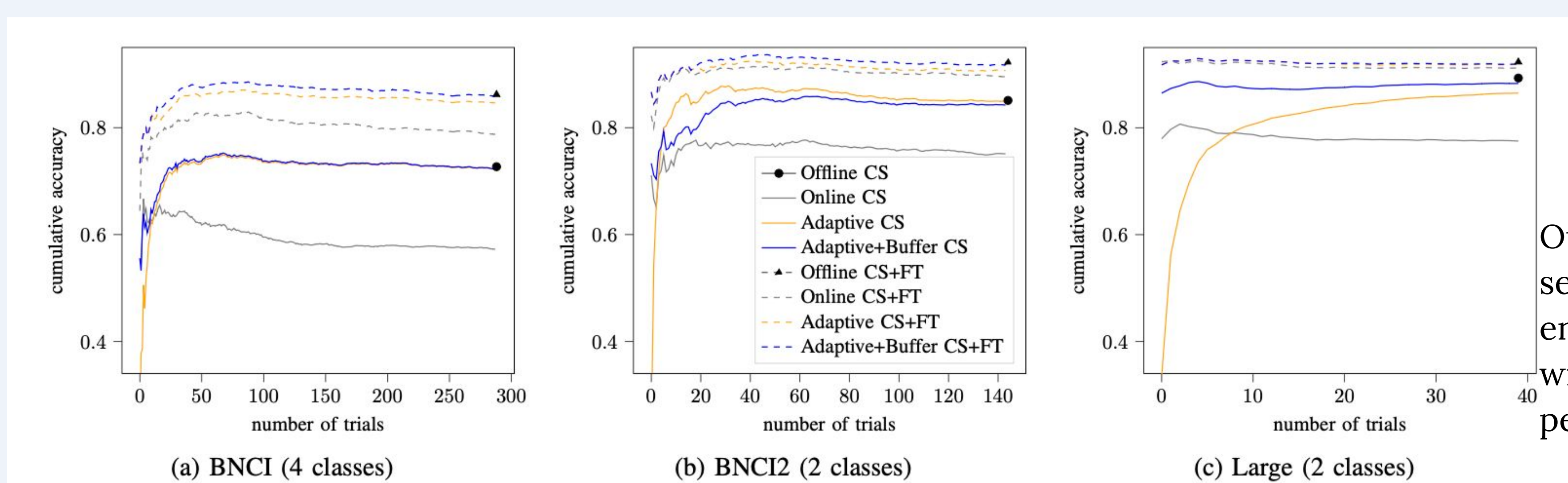


Fig. 1: Adaptive classifier performance compared to online and offline baselines in Cross-Subject (CS) and Cross-Subject Fine-Tuning (CS+FT) evaluations.

Our method enables real-time session-specific adjustments while enhancing model adaptability, without compromising performance.

Conclusion

- ❖ **Adaptive Method:** Proposed a real-time BCI method using deep learning backbones without retraining.
- ❖ **Key Approach:** Updates normalizations at both input and latent spaces.
- ❖ **Outcomes:** The method bridge the gap between online and offline, and is applied to multiple MI datasets and evaluation setups.
- ❖ **Future Work:** Effectiveness in real online studies remains to be tested. Potential application for fast calibration in few-shot setups.

We think it might help designing BCI systems that are easier and faster to calibrate to a new user

References

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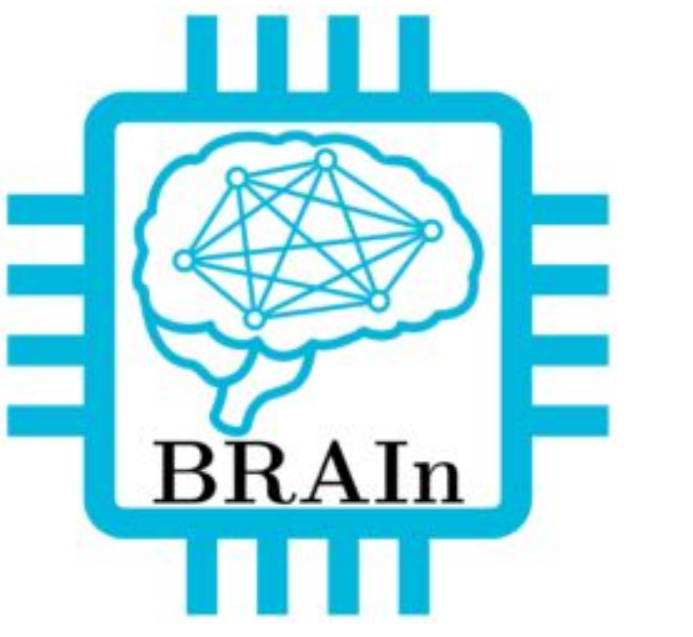


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TL;DR

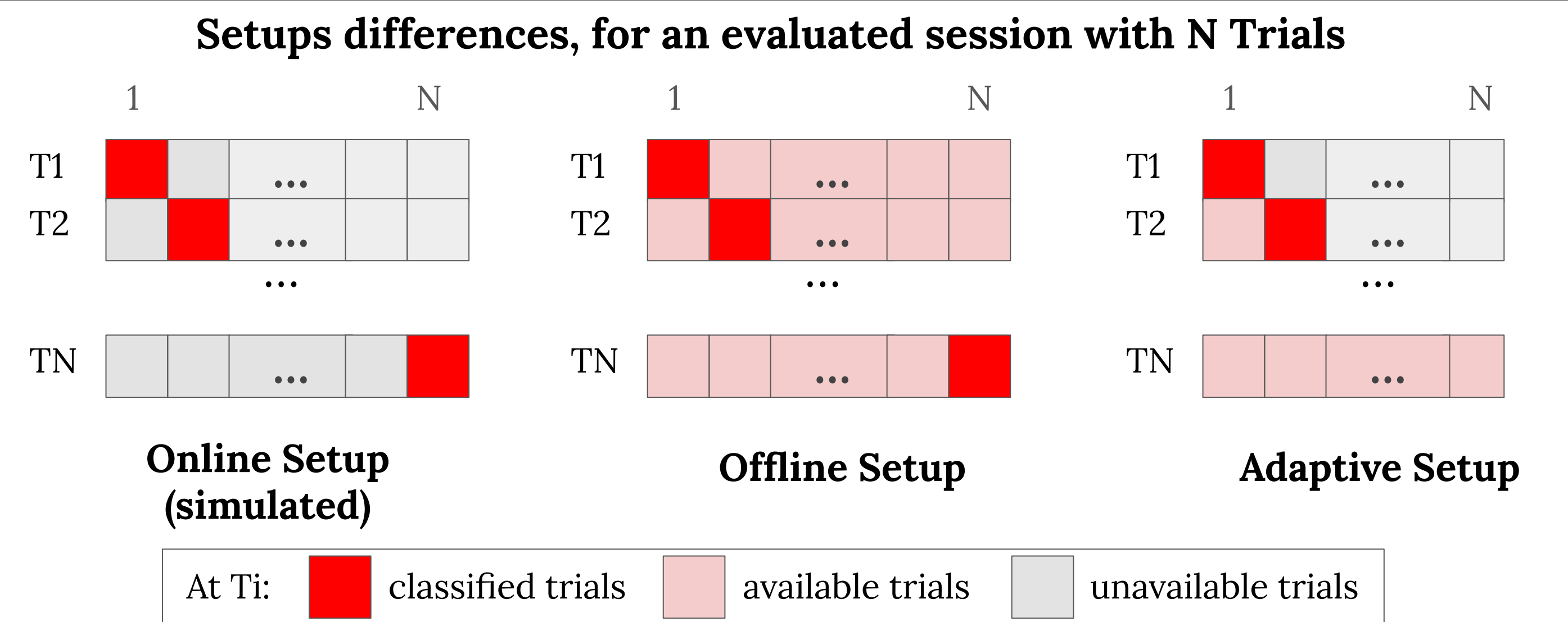
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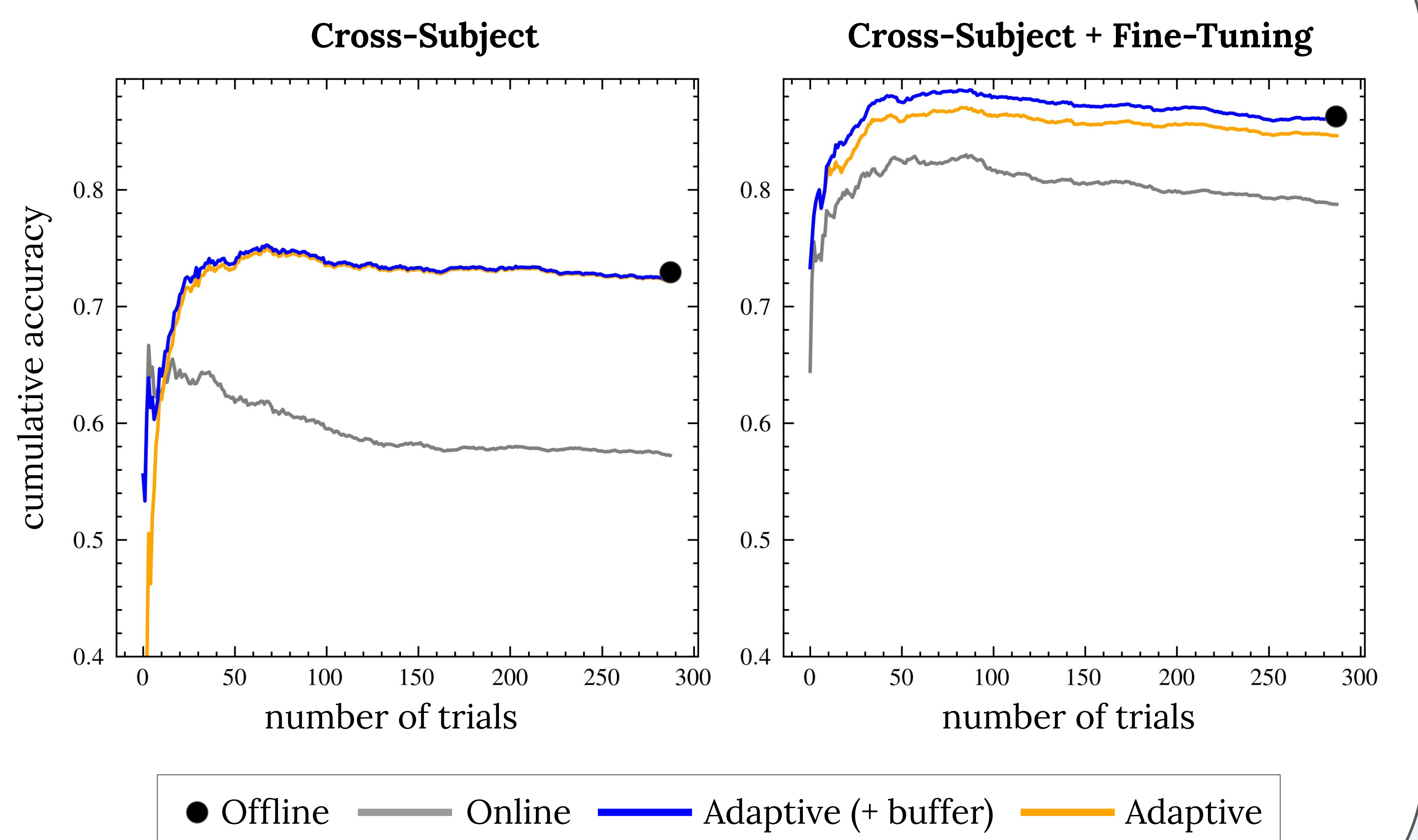
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Performance comparison over the time between the adaptive method (with and without buffer) and the online and offline baselines



Our method enables real-time session-specific adjustments while enhancing model adaptability, without compromising performance.

Conclusion

- ❖ **Adaptive Method:** Proposed a real-time BCI method using deep learning backbones without retraining.
- ❖ **Key Approach:** Updates normalizations at both input and latent spaces.
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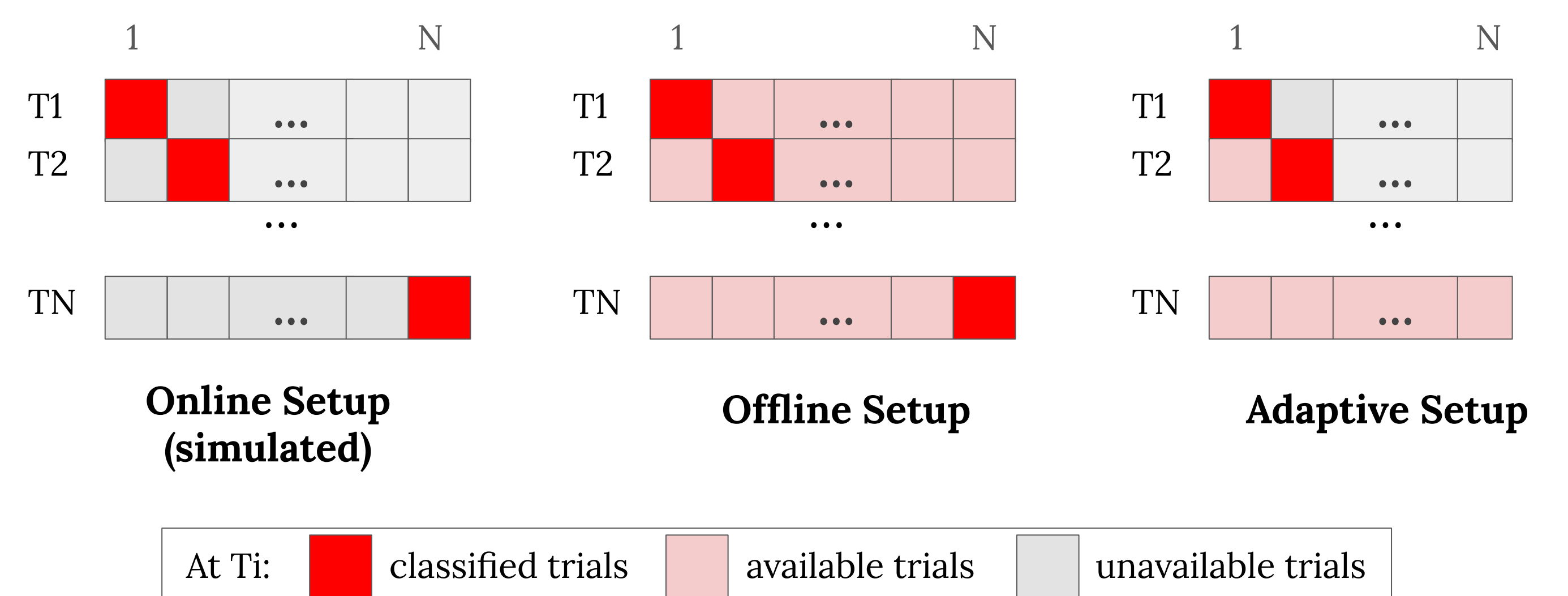
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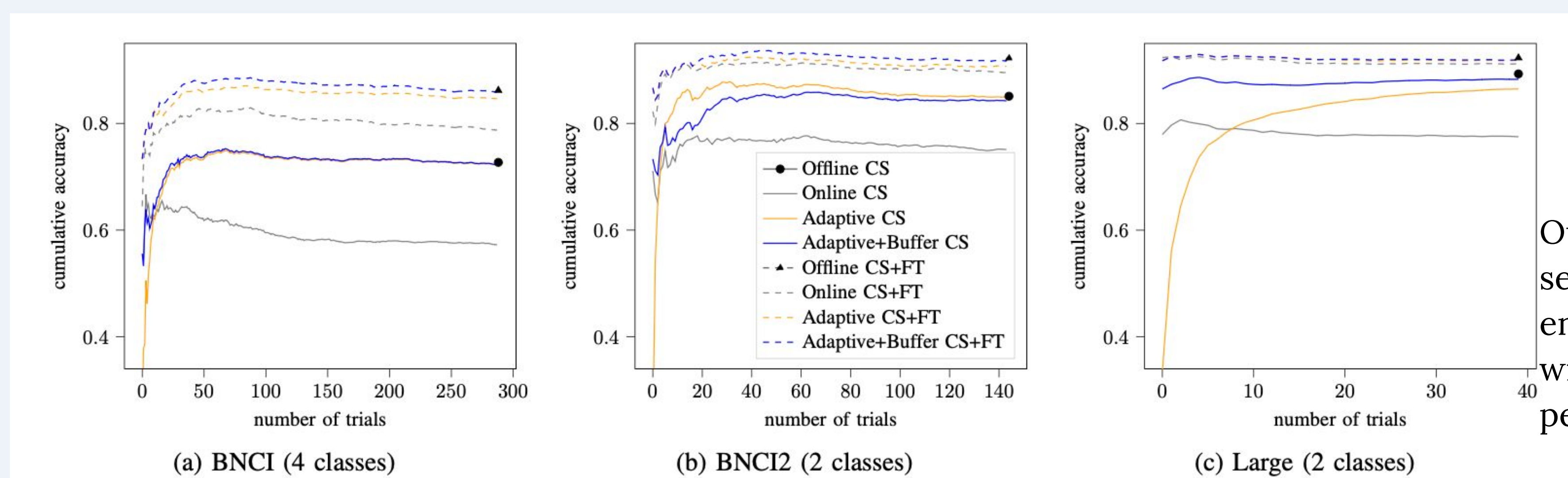


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